21-241 Matrices and Linear Transformations Notes

Compiled by Salman Hajizada

Exam 2 Edition

Chapter 1

Linear Equations in Linear Algebra

1.1 Systems of linear equations

Definition: A system of linear equations is a collection of one or more linear equations involving the same set of variables. A system has either:

- No solution
- Infinite solutions
- A unique solution

The info about a linear system can be recorded in a matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

The one with just the coefficients is called **coefficient matrix**
The one with the last col with the product is called the **augmented matrix**

There are 3 elementary row operations to carry out on the matrix:

- Replacement: by a sum of itself and a multiple of another row
- Interchange: swap rows
- Scaling: scale a row by a scalar

1.2 Row Reduction and Echelon Forms

Definition: A matrix is in **echelon form** if

- All nonzero rows are above any rows of zeros
- Each leading entry of a row is in a column ro the right of the leading entry of the row above it
- All entries in a column below a leading entry are zeros

A matrix is in **reduced echelon form** if

- It is in echelon form
- The leading entry in each nonzero row is 1
- Each leading 1 is the only nonzero entry in its column

Theorem 1 (Uniqueness of the Reduced Echelon Form)**.** *Each matrix is row equivalent to one and only one reduced echelon matrix*

Definition 1 (Pivot Positon)**.** *A **pivot position** in A is a location in A that corresponds to a leading 1 in the reduced echelon form of A*

Row Reduction Algorithm

1. Start with left non-zero column. This is a pivot column. Pivot position is at the top
2. Choose a nonzero entry in the pivot column as a pivot. Interchange to move it into pivot position
3. Use row replacement to make zeros below pivot position
4. Ignore the row with the pivot and the ones above. Apply steps 1-3 to the remaining matrix until no more rows to modify
5. Starting with rightmost pivot and working up and left, make zeros above each pivot. Make every pivot a 1 by scaling

Variables corresponding to pivot columns are **basic**, the other ones are **free**

Theorem 2 (Existence and Uniqueness theorem)**.**

A linear system is consistent iff the right column of the augmented matrix is NOT a pivot column i.e. iff echelon form has NO row of form:

$$\begin{bmatrix} 0 & \dots & 0 & b \end{bmatrix} \text{ with } b \text{ nonzero}$$

If a system is consistent then solution set contain either a unique solution if there are 0 free variables or infinite solutions if there is at least 1 free variable.

1.3 Vector Equations

Algebraic Properties of Vectors in $\mathbb{R}^n$

$\forall\ \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and scalars c and d:

1. **Commutativity of addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. **Associativity of addition:** $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
3. **Additive identity:** $\mathbf{u} + \mathbf{0} = \mathbf{u}$
4. **Additive inverse:** $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
5. **Distributivity of scalar multiplication over addition (vectors):** $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
6. **Distributivity of scalar multiplication over addition (scalars):** $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
7. **Associativity of scalar multiplication:** $(cd)\mathbf{u} = c(d\mathbf{u})$
8. **Scalar identity:** $1\mathbf{u} = \mathbf{u}$

A linear combination is a vector $\mathbf{y} = c_1\mathbf{v}_1 + \ldots + c_p\mathbf{v}_p$

Definition 2. *Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is the set of all linear combinations of $\mathbf{v_1}, \ldots, \mathbf{v_p}$*
It is the set of all vectors that can be written in the form $c_1\mathbf{v}_1 + \ldots + c_p\mathbf{v}_p$

1.4 The Matrix Equation Ax = b

Definition 3.

If A is an $m \times n$ matrix, with columns $\mathbf{a_1}, \ldots, \mathbf{a_n}$ and if $\mathbf{x} \in \mathbb{R}^n$,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

This is defined only if the number of columns of A equals number of entires in $\mathbf{x}$

Theorem 3.

If A is an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$, then $A\mathbf{x} = \mathbf{b}$ has the same solution set as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = b$$

which has the same solution set as the system of linear equations with matrix

$$\begin{bmatrix} \mathbf{a}_1 & \ldots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}$$

Theorem 4.

Let A be a $m \times n$ matrix. The following statements are equivalent

a. $\forall\ \mathbf{b} \in \mathbb{R}^m, A\mathbf{x} = \mathbf{b}$ has a solution

b. $\forall\ \mathbf{b}$ is a linear combination of the columns of A

c. The columns of A span $\mathbb{R}^m$

d. A has a pivot in every row

Warning: A is a **coefficient matrix** in this theorem

Theorem 5.

If A is an $m \times n$ matrix and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, and c is a scalar

a. $A\ (\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$;

b. $A\ (c\mathbf{u}) = c(A\mathbf{u})$;

1.5 Solution sets of Linear Systems

Homogenous Linear Systems: if it can be written in form $A\mathbf{x} = \mathbf{0}$
This system always has at least one solution ($\mathbf{x} = \mathbf{0}$): **trivial solution**
The equation has a **non-trivial** solution iff it has at least one free variable

Nonhomogenous Linear Systems: if it can be written in form $A\mathbf{x} = \mathbf{b}$, $\mathbf{b}$ non-zero

Theorem 6.

Let $A\mathbf{x} = \mathbf{b}$ be consistent for some $\mathbf{b}$, let $\mathbf{p}$ be a solution. Then the solution set is the set of all vectors in form $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$, where $\mathbf{v}_h$ is any solution of $A\mathbf{x} = \mathbf{0}$

Writing a solution set in parametric vector form

1. Row reduce to reduced echelon form
2. Express each basic variable in terms of any free variables
3. Write a typical solution as a vector whose entries depend on free variables
4. Decompose it into a linear combination of vectors (with numbers only inside) using free variables as parameters

1.6 Application of Linear Systems

idk how to make it skip it. So just ignore this

1.7 Linear Independence

Definition 4.

$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is **linearly independent** if

$$x_1\mathbf{v}_1 + \ldots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. It is said to be **linearly dependent** if $\exists c_1, \ldots, c_p$, not all zero, such that

$$c_1\mathbf{v}_1 + \ldots + c_p\mathbf{v}_p = \mathbf{0}$$

A set of two vectors is linearly dependent if at least one is a multiple of the other

Theorem 7 (Characterization of Linearly independent Sets)**.**

A set of two or more vectors is linearly dependent iff at least one of them is a linear combination the others.

Theorem 8.

If a set contains more vectors than entries in each vector, then it is linearly dependant

Theorem 9.

If a set contains the zero vector, then it is linearly dependant

1.8 Introduction to Linear Transformations

A **transformation** T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns each vector $\mathbf{x}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{x})$ in $\mathbb{R}^m$

- Domain: $\mathbb{R}^n$
- Codomain: $\mathbb{R}^m$
- Range: set of all images $T(\mathbf{x})$

Definition 5. *T is linear iff*

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
- $T(c\mathbf{u}) = cT(\mathbf{u})$

If T is linear then:

- $T(\mathbf{0}) = \mathbf{0}$
- $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$
- $T(c_1\mathbf{v}_1 + \ldots + c_p\mathbf{v}_p) = T(c_1\mathbf{v}_1) + \ldots + T(c_p\mathbf{v}_p)$

1.9 The Matrix of a Linear Transformation

Theorem 10.

For T, there exists a unique A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

Where $A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \ldots & T(\mathbf{e}_n) \end{bmatrix}$
With $\mathbf{e}_j$ is the jth column of the identity matrix

Definition 6.

T is **onto** (surjective) if

- Range of T = Codomain
- $\forall\ \mathbf{v} \in codomain(B), \exists\ \mathbf{u} \in domain(A)$ such that $T(\mathbf{u}) = \mathbf{v}$
- Find standard matrix
$$A = \begin{bmatrix} T(c_1) & \dots & T(c_n) \end{bmatrix}$$
Onto if $A\mathbf{u} = \mathbf{v}$ has a solution for each $\mathbf{v} \in Codomain$

Definition 7.

T is **one-to-one** (injective)if

- $T(\mathbf{u}_1) = T(\mathbf{u}_2) \implies \mathbf{u}_1 = \mathbf{u}_2$
- Find standard matrix
$$A = \begin{bmatrix} T(c_1) & \dots & T(c_n) \end{bmatrix}$$
$A\mathbf{u} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in Range(T)$
Or
$A\mathbf{u} = \mathbf{b}$ has at most one solution for each $\mathbf{b} \in Codomain$ — (can have no solution)

Theorem 11. *T is one-to-one iff $T(\mathbf{x}) = \mathbf{0}$ has a ONLY the trivial solution*

Theorem 12. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$, let A be standard matrix for T. Then*

- *T maps $\mathbb{R}^n$ to $\mathbb{R}^m$ iff the columns of A span $\mathbb{R}^m$*
- *T is one-to-one iff the columns of A are linearly independent*

Chapter 2

Matrix Algebra

2.1 Matrix Operations

- **Square Matrix**: row and columns are the same
- **Diagonal entries**: $a_{11}, a_{22}, \ldots$
- **Diagonal matrix**: square $n \times n$ matrix who's non-diagonal entires are zero
- **Zero matrix**: $m \times n$ matrix where all entries are 0

$[a_{ij}]_{m\times n} \pm [b_{ij}]_{m\times n} = [a_{ij} \pm b_{ij}]_{m\times n}$

Two matrices are **equal** if they have the same size and their entries are equal (duh)

Theorem 1.

Properties of Matrices

Let A, B and C be matrices of the same size, and let r and s be scalars

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0 = A$
- $r(A + B) = rA + rB$
- $(r + s)A = rA + sA$
- $r(sA) = (rs)A$

Definition 1. *If A is a $m \times n$ matrix and B is an $n \times p$ matrix with columns $\mathbf{b}_1, \ldots, \mathbf{b}_p$, then*

$$AB = A\left[\mathbf{b}_1 \ldots \mathbf{b}_p\right] = \left[A\mathbf{b}_1 \ldots A\mathbf{b}_p\right]$$

- Each column of AB is a linear combination of the columns of A using weights from columns of B
- **The number of columns of A must match the number of rows in B**
- Row column rule: $(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}$
- $row_i(AB) = row_i(A) \cdot B$

Theorem 2.

Properties of Matrices

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.

$A(BC) = (AB)C$ (associative law of multiplication)

$A(B + C) = AB + AC$ (left distributive law)

$(B + C)A = BA + CA$ (right distributive law)

$r(AB) = (rA)B = A(rB)$ for any scalar r

$I_m A = A = AI_n$ (identity for matrix multiplication)

WARNING

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.

- In general, $AB \neq BA$
- In general, $AB = AC$ **does not imply** $B = C$
- In general, $AB = 0$ **does not imply** $A = 0 \vee B = 0$

Power of a matrix: $A^k = \underbrace{AA \dots A}_{k}$

The transpose of a matrix: A^T is obtained by swapping its rows and columns of A.

Theorem 3.

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(rA)^T = rA^T$
- $(AB)^T = B^T A^T$

2.2 The Inverse of a Matrix

An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$CA = I_n \quad \text{and} \quad AC = I_n,$$

In this case, C is an **inverse** of A

$$A^{-1}A = I \text{ and } AA^{-1} = I$$

Theorem 4. *Let* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *If* $ad - bc \neq 0$, *then* A *is invertEBIL and*

$$A^-1 = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$detA = ad - bc$

Theorem 5. *If* A *is invertible, then* $\forall\ \mathbf{b} \in \mathbb{R}^n$, $A\mathbf{x} = \mathbf{b}$ *has the unique solution* $\mathbf{x} = A^{-1}\mathbf{b}$

Theorem 6. *1. If A invertible, then A^{-1} too and $(A^{-1})^{-1} = A$*

2. If A and B invertible, $(AB)^{-1} = B^{-1}A^{-1}$

3. If A invertible, $(A^T)^{-1} = (A^{-1})^T$

- An **elementary matrix** is a matrix obtained by performing a single elementary row operation on I
- If a row op if performed on A, the result can be written as EA, where E is made by performing the same row op on I
- Each E is an invertible and its inverse is the elementary matrix that transforms E back into I.

Theorem 7. *A is invertible iff A is row equivalent to I, and any sequence of row ops that reduces A to I also transforms I to A^{-1}*

Finding A^{-1}: Row reduce $\begin{bmatrix} A & I \end{bmatrix}$. If A row-eq to I then $\begin{bmatrix} A & I \end{bmatrix}$is row-eq to $\begin{bmatrix} I & A^{-1} \end{bmatrix}$

2.3 Characterizations of Matrices

Theorem 8.

The following statements about a square n by n matrix A are equivalent

- A is an invertible matrix.
- A is row equivalent to the $n \times n$ identity matrix.
- A has n pivot positions.
- The equation $Ax = 0$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $x \mapsto Ax$ is one-to-one.
- The equation $Ax = b$ has at least one solution for each $b \in \mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $x \mapsto Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

If $AB = I$, then A and B invertible with $B = A^{-1}$ and $A = B^{-1}$
A linear transformation is invertible if $\exists S : \mathbb{R}^n \to \mathbb{R}^n$ such that:

$$\begin{aligned} S(T(\mathbf{x})) &= x \quad \text{for all } \mathbf{x} \in \mathbb{R}^n \\ T(S(\mathbf{x})) &= x \quad \text{for all } \mathbf{x} \in \mathbb{R}^n \end{aligned}$$

Theorem 9. *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ and A be the matrix. T is invertible iff A is invertible. Then $S(\mathbf{x}) = A^{-1}\mathbf{x}$*

2.4 Partitioned Matrices

Trivial stuff honestly just partition and multiply.
Ok actually there is one good theorem here

Theorem 10 (Column-Row Expansion of AB)**.**

If A is $m \times n$ and B is $n \times p$:

$$AB = \begin{bmatrix} col_1(A) & \dots & col_n(A) \end{bmatrix} \begin{bmatrix} row_1(B) \\ \vdots \\ row_n(B) \end{bmatrix} = col_1(A)row_1(B) + \dots + col_n(A)row_n(B)$$

The inverse of a block upper triangular matrix:

$$A^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

2.5 Matrix Factorizations

Assume A is $m \times n$ and can be reduced to Echelon Form without row interchanges.
Then $A = LU$,

- $L : m \times m$ lower triangular with 1's in diagonal
- $U : m \times n$ echelon form of A

$A\mathbf{x} = \mathbf{b}$ can be written as $L(U\mathbf{x}) = \mathbf{b}$. Writing $\mathbf{y} = U\mathbf{x}$, we get:

$$L(U\mathbf{x}) = \mathbf{b}$$

$$\begin{aligned} L\mathbf{y} &= \mathbf{b} \\ U\mathbf{x} &= \mathbf{y} \end{aligned}$$

We can get $\mathbf{x}$ by solving the easy pair of equations:

1. Solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$
2. Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$

LU Factorization Algorithm (boring)

1. Reduce A to echelon form U using a sequence of row replacement operations
2. Place entries in L such that the same sequence of row operations reduces L to I

LU Factorization Algorithm (fast)

For a matrix $A : m \times n$

1. Initialize L as the identity matrix $m \times m$ and U as A.
2. For each row $i = 1, 2, \dots, m$:
   - Use row operations to create zeros below the diagonal in U by subtracting multiples of the current row from the rows below.
   - For each row operation, divide the corresponding entries below the pivot in column i by the pivot, and store these multipliers in the corresponding positions of L.
3. Continue this process until U is in echelon form and L has recorded the multipliers such that $A = LU$.

This description is very bad, please practice this on a non-square matrix

Chapter 3

Determinants

3.1 Intro to Determinants

First, we denote A_{ij} to be the sub-matrix of A formed by deleting the ith row and jth column of A

Definition 1. *For $n \geq 2$, the **determinant** of a $n \times n$:*

$$\begin{aligned} \det A &= a_{11}\det(A_{11}) - a_{12}\det(A_{12}) + \ldots + (-1)^{1+n}a_{1n}\det(A_{1n}) = \\ &= \textstyle\sum_{j=1}^{n}(-1)^{1+j}a_{1j}\det(A_{1j}) \end{aligned}$$

The (i, j)-cofactor of A: $C_{ij} = (-1)^{i+j}\det(A_{ij})$

Theorem 1. *The determinant of A can be computed by cofactor expansion across any row or column.*
Across row i:

$$\det(A) = a_{i1}C_{i1} + \ldots + a_{in}C_{in}$$

Across column j:

$$\det(A) = a_{1j}C_{1j} + \ldots + a_{nj}C_{nj}$$

Signs of the cofactor:

$$\begin{bmatrix} + & - & + & \ldots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

Theorem 2. *If A triangular, then $\det(A)$ is the product of the entires on the main diagonal of A*

3.2 Properties of Determinants

Theorem 3.

Row Operations

For a matrix $A : m \times n$

a. If a multiple of a row in A is added to another to produce B: $\det(B) = \det(A)$

b. If two rows of A are swapped to produce B: $\det(B) = -\det(A)$

c. If a row of A is multiplied by k to produce B: $\det(B) = k \cdot \det(A)$

Two main strats where this is useful:

- Row reduce a matrix to echelon form to simplify calculation (without row interchange!)
- Factor out a common factor from a row to simplify calculation

Theorem 4. *A is invertible iff* $\det(A) \neq 0$ *(trivial)*

Theorem 5. $\det(A) = \det(A^T)$

Theorem 6. *If A and B $n \times n$, then* $\det(AB) = \det(A) \cdot \det(B)$

Lemma 1. $\det(A^{-1}) = \frac{1}{\det(A)}$

3.3 Crammers Rule

Let $A_i(\mathbf{b})$ be the matrix obtained from A by replacing column i by the vector $\mathbf{b}$

Theorem 7 (Cramer's Rule)**.** *Let $A : n \times n$, invertible, the solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ is given by:*

$$x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}, \text{ for } i = 1, 2, \ldots, n$$

The matrix of cofactors is called the **adjugate or classical adjoint** of A, denoted by adj A

Theorem 8 (an Inverse Forumla)**.**

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} C_{11} & C_{21} & \ldots & C_{n1} \\ C_{12} & C_{22} & \ldots & C_{n2} \\ \vdots & \vdots & & \\ C_{1n} & C_{2n} & \ldots & C_{nn} \end{bmatrix} = \frac{adj(A)}{\det(A)}$$

Theorem 9. *If $A : 2 \times 2$, Area of parallelogram determined by columns of A is* $|\det(A)$
If $A : 3 \times 3$, Volume of parallelepiped determined by columns of A is $|\det(A)$

Chapter 4

Vector Spaces and Subspaces

4.1 Vector Spaces and Subspaces

Definition 2. *A **vector space** is a nonempty set V of objects called vectors, on which there are defined two operations: addition and scalar multiplication. The axioms below must hold for all $\mathbf{u}, \mathbf{w}, \mathbf{v} \in V$*

1. $\mathbf{u} + \mathbf{v} \in V$ (Closure under addition)
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. $\mathbf{0} \in V$, such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$
5. $\forall\ \mathbf{u} \in V,\ \exists\ -\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
6. $c\mathbf{u} \in V$
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$
10. $1\mathbf{u} = \mathbf{u}$

Definition 3. *A **subspace** of a space V is a subset H of V that:*

- *The zero vector of V is in H*
- *H is closed under vector addition: $\mathbf{u} + \mathbf{v} \in H$ for all $\mathbf{u}, \mathbf{v} \in H$*
- *H is closed under scalar multiplication: $c\mathbf{u} \in H$ for all scalars c and vectors $\mathbf{u} \in H$*

Theorem 10. *If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in a vector space V, then $Span\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a subspace of V*